

Supplementary Material to “Towards Standardized Evaluation of Feasible Region Identification in Constrained Engineering Design”

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APPENDIX A

BENCHMARK PROBLEM DESCRIPTIONS AND DEFINITIONS

A. Six-Hump Camel

A classic optimization test function that is set up as a single non-linear constraint, resulting in a single non-convex feasible region. Despite the relatively large feasible region, the difficulty lies in its nonconvex shape, due to the narrow part in the middle of the design space that might be considered infeasible by some algorithms. The constraint function is [1]:

$$g_1(\mathbf{x}) = (4 - 2.1x_1^2 + \frac{x_1^4}{3})(x_1^2) + x_1x_2 + (-4 + 4x_2^2)(x_2^2) - 0.1 \leq 0,$$

with $x_1, x_2 \in [-1, 1]$.

B. Hosaki

This constraint function [1], [2] creates two disconnected feasible regions, where one is much larger than the other, therefore algorithms might overlook the small region and only focus on the larger one.

$$g_1(\mathbf{x}) = \left(1 - 8x_1 + 7x_1^2 - \frac{7}{3}x_1^3 + \frac{1}{4}x_1^4\right)x_2^2 \exp(-x_2) + 1.1 \leq 0$$

The design space is $x_1 \in [0, 5], x_2 \in [0, 6]$.

C. Sinexp

As the constraint function consists of an exponential periodic function, it is difficult to model. Depending on the choice of threshold value, the two feasible regions can be narrower or wider making it harder or easier, respectively, to find.

$$g_1(\mathbf{x}) = x_1 \sin(5\pi x_2) + \exp(x_2 - 0.5) - 0.501 \leq 0$$

The design space is $x_1, x_2 \in [0, 1]$.

D. Branin

This function has been used in various FRI papers [2], [3] because it consists of three disconnected regions that take up less than 5% of the design space. Also here the difficulty lies in identifying all three regions.

$$g_1(\mathbf{x}) = \left(x_2 - \frac{5.1}{4\pi^2}x_1^2 + \frac{5}{\pi}x_1 - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos(x_1) + 10 - 8 \leq 0$$

The design space is $x_1 \in [-13, 14], x_2 \in [-8, 23]$.

E. Himmelblau

Another common optimization function [4] used as a constraint. We have chosen the constraint threshold so that three feasible regions are created: two convex round-shaped regions and one larger slightly nonconvex. With this setting it is again important to make sure the algorithm has found all three regions. The single constraint function is:

$$g_1(\mathbf{x}) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2 - 20 \leq 0,$$

where $x_1, x_2 \in [-5, 5]$.

F. Donut

We created a donut shaped feasible region problem using one constraint function with a lower and upper threshold. With this problem, it is interesting to see how much sampling is spent in the hole of the donut.

$$g_1(\mathbf{x}) = \exp(-2(x_1 - 0.5)^2 - 4(x_2 - 0.6)^2) - 0.75 \leq 0$$

$$g_2(\mathbf{x}) = -(\exp(-2(x_1 - 0.5)^2 - 4(x_2 - 0.6)^2) - 0.55) \leq 0$$

The design space is $x_1, x_2 \in [0, 1]$.

G. Sasena

This problem [1] consists of three constraints that, when combined, result in one smaller and one larger region. The three constraints are:

$$\begin{aligned} g_1(\mathbf{x}) &= (x_1 - 3.0)^2 + (x_2 + 2)^2 \exp(-x_2^7) - 12 \leq 0 \\ g_2(\mathbf{x}) &= 10x_1 + x_2 - 7 \leq 0 \\ g_3(\mathbf{x}) &= (x_1 - 0.5)^2 + (x_2 - 0.5)^2 - 0.2 \leq 0, \end{aligned}$$

where $x_1, x_2 \in [0, 1]$.

H. Cheese

This six-constraint problem [5] creates a very interesting feasible region. Five out of six constraints are actively constraining. The feasible region is nonconvex due to the hole and cutout by two constraints.

$$\begin{aligned} g_1(\mathbf{x}) &= x_2 - 2x_1 - 15 \leq 0 \\ g_2(\mathbf{x}) &= \frac{x_1^2}{2} + 4x_1 - 5 - x_2 \leq 0 \\ g_3(\mathbf{x}) &= 10 - \frac{(x_1 - 4)^2}{5} - \frac{x_2^2}{0.5} \leq 0 \\ g_4(\mathbf{x}) &= x_2 - 15 \leq 0 \\ g_5(\mathbf{x}) &= x_2(6 + x_1) - 80 \leq 0 \\ g_6(\mathbf{x}) &= -(x_1 + 5)^2 - (x_2 + 5)^2 + 2 \leq 0 \end{aligned}$$

The design space is $x_1 \in [-10, 5], x_2 \in [-15, 15]$.

I. Unequal Spheres

As we want to make sure algorithms can find multiple feasible regions in higher dimensions, we created this 3D example that consists of five feasible spheres with different radii. While the problem consists of only one constraint, algorithms are not always able to find all five regions as some are rather small.

The union of m discs in d dimensions is defined as:

$$g_1(\mathbf{x}) = \min_{i=1}^m \|\mathbf{x} - \mathbf{c}_i\|_2 - r_i \leq 0 \quad (1)$$

where $\mathbf{c}_i$ is the center and r_i the radius of disc i . In our experiments, we use 5 discs in 3D with

$$\begin{aligned} \mathbf{c} &= [(0.2, 0.2, 0.2), (0.8, 0.2, 0.5), \\ &\quad (0.8, 0.8, 0.8), (0.2, 0.8, 0.5), (0.5, 0.5, 0.5)] \\ r &= [0.05, 0.1, 0.2, 0.05, 0.1], \end{aligned}$$

The design space is $[0, 1]^d$.

J. Equal Spheres

This 5D variant of the previous problem consists of five feasible regions of equal size. Nevertheless, the difficulty of finding all of them remains due to the higher dimensional space. It is a good way to show how algorithms deal with the exploration-exploitation trade-off, as in this case exploring more might actually be beneficial to finding all regions.

The constraint function is the same as for Unequal Spheres (see Eq. 1), however we now consider 5 equal discs in 5D with

$$\begin{aligned} \mathbf{c} &= [(0.2, 0.2, 0.2, 0.2, 0.2), (0.8, 0.2, 0.5, 0.5, 0.2), \\ &\quad (0.8, 0.8, 0.8, 0.8, 0.8), (0.2, 0.8, 0.5, 0.5, 0.8), \\ &\quad (0.5, 0.5, 0.5, 0.5, 0.5)] \\ r &= [0.18, 0.18, 0.18, 0.18, 0.18]. \end{aligned}$$

The design space is again $x_1, x_2, x_3, x_4, x_5 \in [0, 1]$.

K. Crescent

A crescent shaped feasible region is an interesting use case also found in constrained optimization benchmarks [6]. What we tried to do here is to take this to a higher dimension. We wanted to create a high-dimensional problem with a known non-convex shaped feasible region. Besides it being a small region, it is its shape that makes it harder to correctly identify.

$$\begin{aligned} g_1(\mathbf{x}) &= \sum_{i=1}^d (x_i - \mathbf{c}_{1,i})^2 - 4.0 \leq 0, \\ g_2(\mathbf{x}) &= 4.0 - \sum_{i=1}^d (x_i - \mathbf{c}_{2,i})^2 \leq 0, \end{aligned}$$

with

$$\begin{aligned} \mathbf{c}_1 &= [1.0, 2.5, 2.5, 2.5, 2.5, 2.5], \\ \mathbf{c}_2 &= [0.0, 2.5, 2.5, 2.5, 2.5, 2.5], \end{aligned}$$

and $x_i \in [0, 6]$.

L. Nowacki Beam

This engineering benchmark problem [3] is created using six constraints. Although not all six constraints are actually boundaries of the feasible region, its nonconvex shape makes this problem very suitable for assessing FRI algorithms.

$$\begin{aligned} g_1(\mathbf{x}) &= x_1 x_2 - 0.0025 \leq 0, \\ g_2(\mathbf{x}) &= \frac{P \lambda^3}{3EI_y} - 0.005 \leq 0, \\ g_3(\mathbf{x}) &= \frac{6P\lambda}{x_1 x_2^2} - \sigma_y \leq 0, \\ g_4(\mathbf{x}) &= \frac{1.5P}{x_1 x_2} - \frac{\sigma_y}{2} \leq 0, \\ g_5(\mathbf{x}) &= 2P - 4/\lambda^2 \sqrt{\frac{GI_T EI_z}{1 - \nu^2}} \leq 0, \\ g_6(\mathbf{x}) &= \frac{x_2}{x_1} - 10 \leq 0, \end{aligned} \quad (2)$$

with

$$I_y = \frac{x_1 x_2^3}{12}, \quad I_z = \frac{x_1^3 x_2}{12}, \quad I_T = I_y + I_z,$$

$$\begin{aligned} \lambda &= 0.5, \quad P = 5000, \quad \sigma_y = 240e6, \quad E = 216.62e9, \\ G &= 86.65e9, \quad \nu = 0.27 \end{aligned}$$

$x_1 \in [0.005, 0.1], x_2 \in [0.02, 0.25]$

M. Tension/Compression Spring

This engineering problem was adapted by [7] to create an interesting use case for FRI. The three constraints define one feasible region as follows:

$$\begin{aligned} g_1(\mathbf{x}) &= \frac{4x_2^2 - x_1x_2}{12566(x_2x_1^3 - x_1^4)} + \frac{1}{5108x_1^2} - 1 \leq 0, \\ g_2(\mathbf{x}) &= 1 - \frac{140.45x_1}{x_2^2x_3} \leq 0, \\ g_3(\mathbf{x}) &= \frac{x_1 + x_2}{1.5} - 1 \leq 0, \end{aligned} \quad (3)$$

where $0.05 \leq x_1 \leq 2$, $0.25 \leq x_2 \leq 1.3$ and $2 \leq x_3 \leq 15$.

N. Speed Reducer

The original constrained optimization problem [8] has been modified before for FRI [7], [9] to use seven constraint, however, we choose to use all eleven original constraints as it creates a very small feasible region, while also adding complexity to the algorithm due to the many constraint functions, which are:

$$\begin{aligned} g_1(\mathbf{x}) &= 27 - x_1x_2^2x_3 \leq 0, \\ g_2(\mathbf{x}) &= 1.93 - \frac{x_2x_6^4x_3}{x_4^3} \leq 0, \\ g_3(\mathbf{x}) &= 1.93 - \frac{x_2x_7^4x_3}{x_5^3} \leq 0, \\ g_4(\mathbf{x}) &= x_2x_3 - 40 \leq 0, \\ g_5(\mathbf{x}) &= \frac{x_1}{x_2} - 12 \leq 0, \\ g_6(\mathbf{x}) &= 1.5x_6 - x_4 + 1.9 \leq 0, \\ g_7(\mathbf{x}) &= 1.1x_7 - x_5 + 1.9 \leq 0, \\ g_8(\mathbf{x}) &= 397.5 - x_1x_2^2x_3^2 \leq 0, \\ g_9(\mathbf{x}) &= \frac{10}{x_6^3} \sqrt{\left(\frac{745x_4}{x_2x_3}\right)^2 + 1.69e7} - 1100 \leq 0, \\ g_{10}(\mathbf{x}) &= \frac{10}{x_7^3} \sqrt{\left(\frac{745x_5}{x_2x_3}\right)^2 + 1.575e8} - 850 \leq 0, \\ g_{11}(\mathbf{x}) &= 5 - \frac{x_1}{x_2} \leq 0, \end{aligned} \quad (4)$$

where $2.6 \leq x_1 \leq 3.6$, $0.7 \leq x_2 \leq 0.8$, $17 \leq x_3 \leq 28$, $7.3 \leq x_4, x_5 \leq 8.3$, $2.9 \leq x_6 \leq 3.9$, and $5 \leq x_7 \leq 5.5$.

APPENDIX B EXPERIMENTAL SETUP

We use Python, more specifically *BoTorch* [10], to create the AL framework and to implement the acquisition functions. The noise-free GPs are constructed using *GpyTorch* [11] with a Matérn 5/2 kernel and priors set as described in Hvarfner et al. [12]. Depending on the problem complexity in terms of input dimension d , number of constraints L and the approximated feasibility ratio ρ , the number of initial designs N and the number of iterations q is chosen as given in Table I. The initial designs are constructed using a Sobol sequence [13], while at every iteration a single design is acquired after optimizing the acquisition function until the maximum number of iterations

is reached. The acquisition functions are all optimized using the default L-BFGS-B algorithm using 10 restarts chosen from 512 Sobol samples. Each experiment is repeated 10 times with a different random seed.

TABLE I: Overview of the experimental variables of the benchmark problems.

Name	d	L	$\rho(\%)$	N	q
Six-Hump Camel	2	1	41.67	5	40
Hosaki	2	1	20.23	5	40
Sinexp	2	1	10.59	5	40
Branin	2	1	4.13	5	40
Himmelblau	2	1	9.42	5	40
Donut	2	2	32.45	5	40
Sasena	2	3	13.61	5	40
Cheese	2	6	32.51	5	40
Unequal Spheres	3	1	4.24	30	120
Equal Spheres	5	1	0.50	50	250
Crescent	6	2	0.36	60	300
Nowacki Beam	2	6	8.62	5	40
Tension/Compression Spring	3	3	33.54	10	60
Speed Reducer	7	11	0.11	20	70

APPENDIX C RESULTS ON ALL BENCHMARK PROBLEMS

We have added all result figures of the problems that were not in the main paper here.

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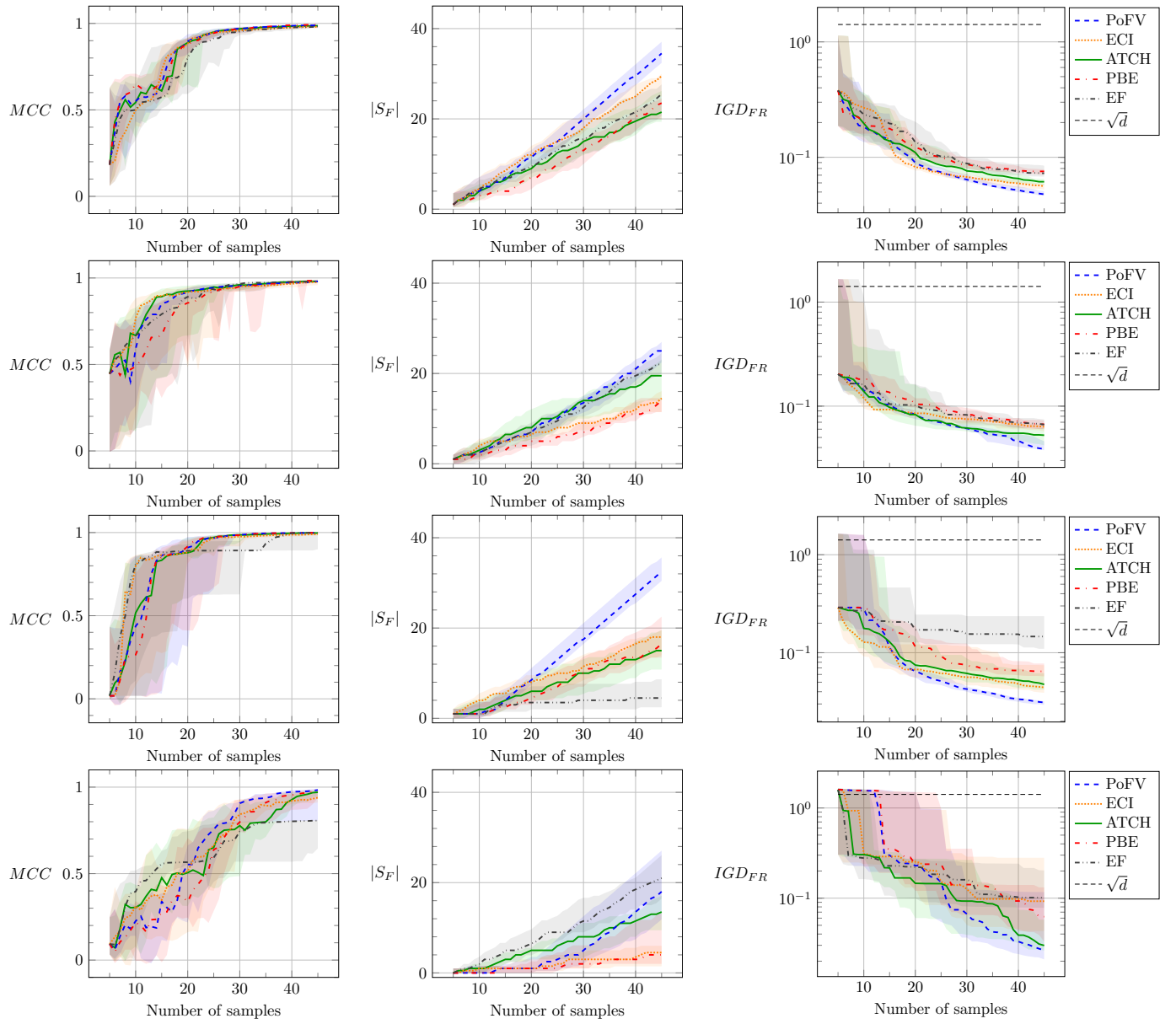


Fig. 1: Performance of the different acquisition functions on the (row 1) T1, (row 2) T2, (row 3) T3, (row 4) T4 problems: (left) MCC performance, (middle) number of feasible designs in the data set, (right) FR-IGD performance. The median and the 5th-95th percentile interval are shown.

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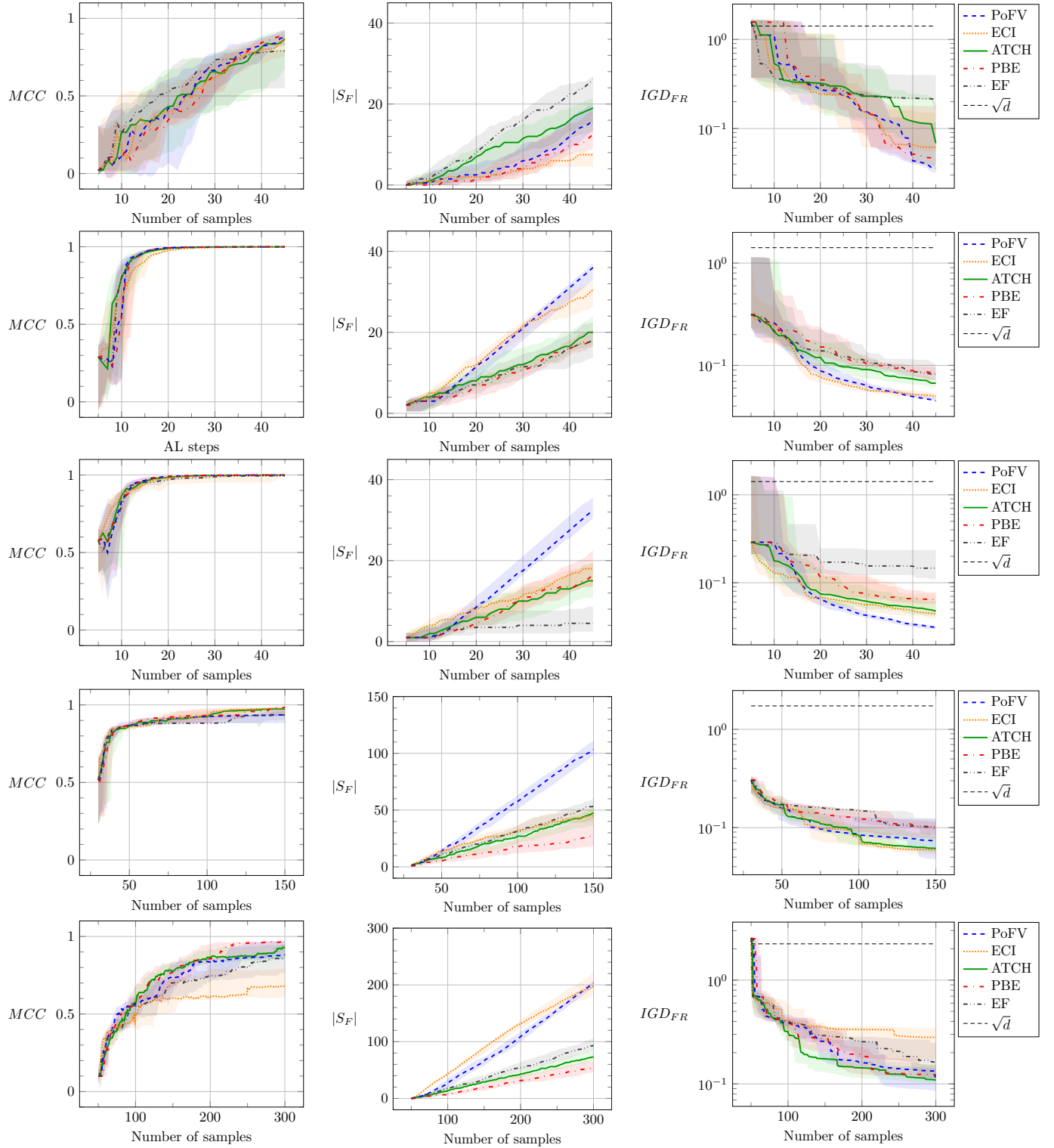


Fig. 2: Performance of the different acquisition functions on the (row 1) T5, (row 2) T6, (row 3) T7, (row 4) T9, (row 5) T10 problems: (left) MCC performance, (middle) number of feasible designs in the data set, (right) FR-IGD performance. The median and the 5th-95th percentile interval are shown.

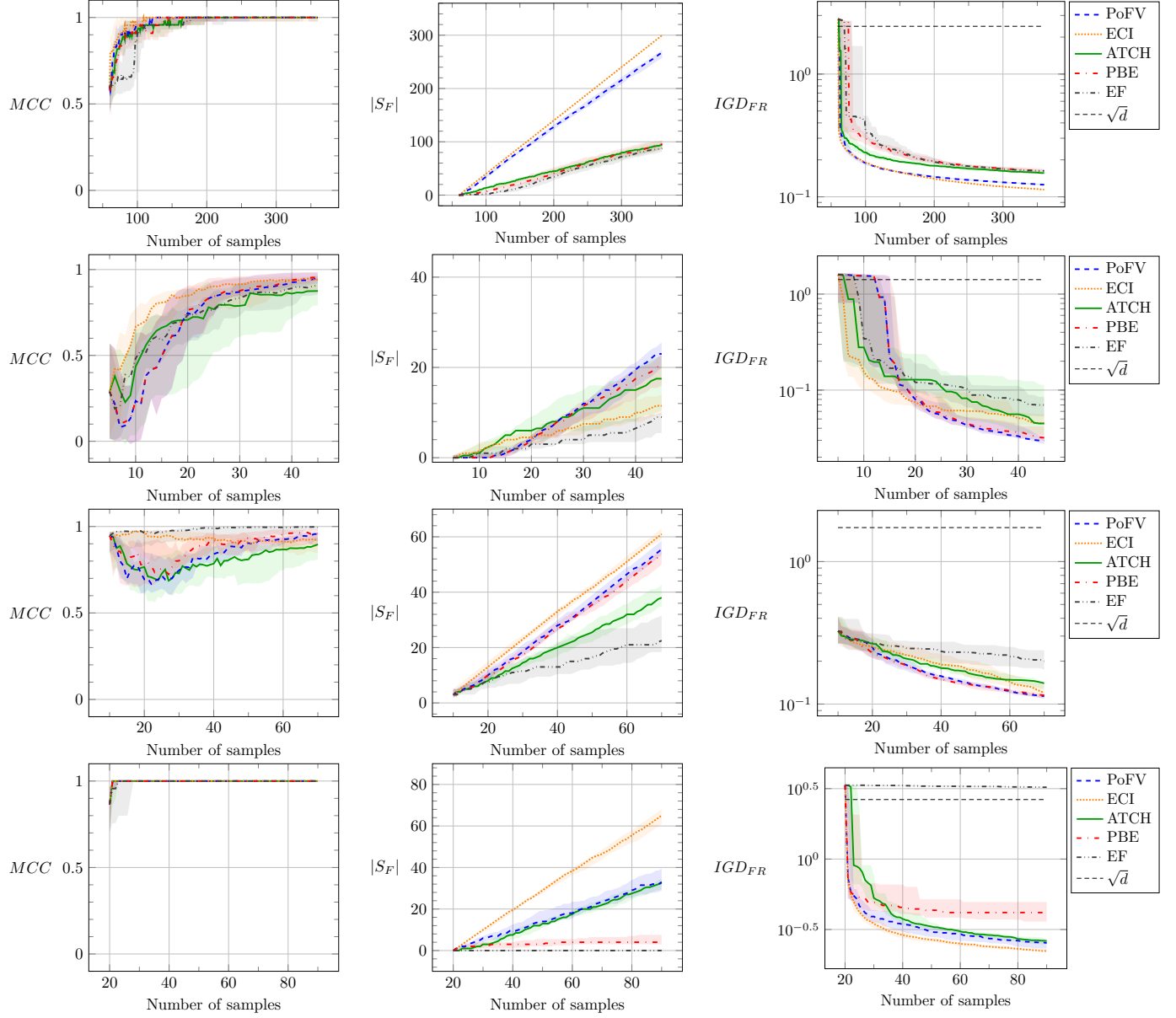


Fig. 3: Performance of the different acquisition functions on the (row 1) T11, (row 2) E1, (row 3) E2, (row 4) E3 problems: (left) MCC performance, (middle) number of feasible designs in the data set, (right) FR-IGD performance. The median and the 5th-95th percentile interval are shown.

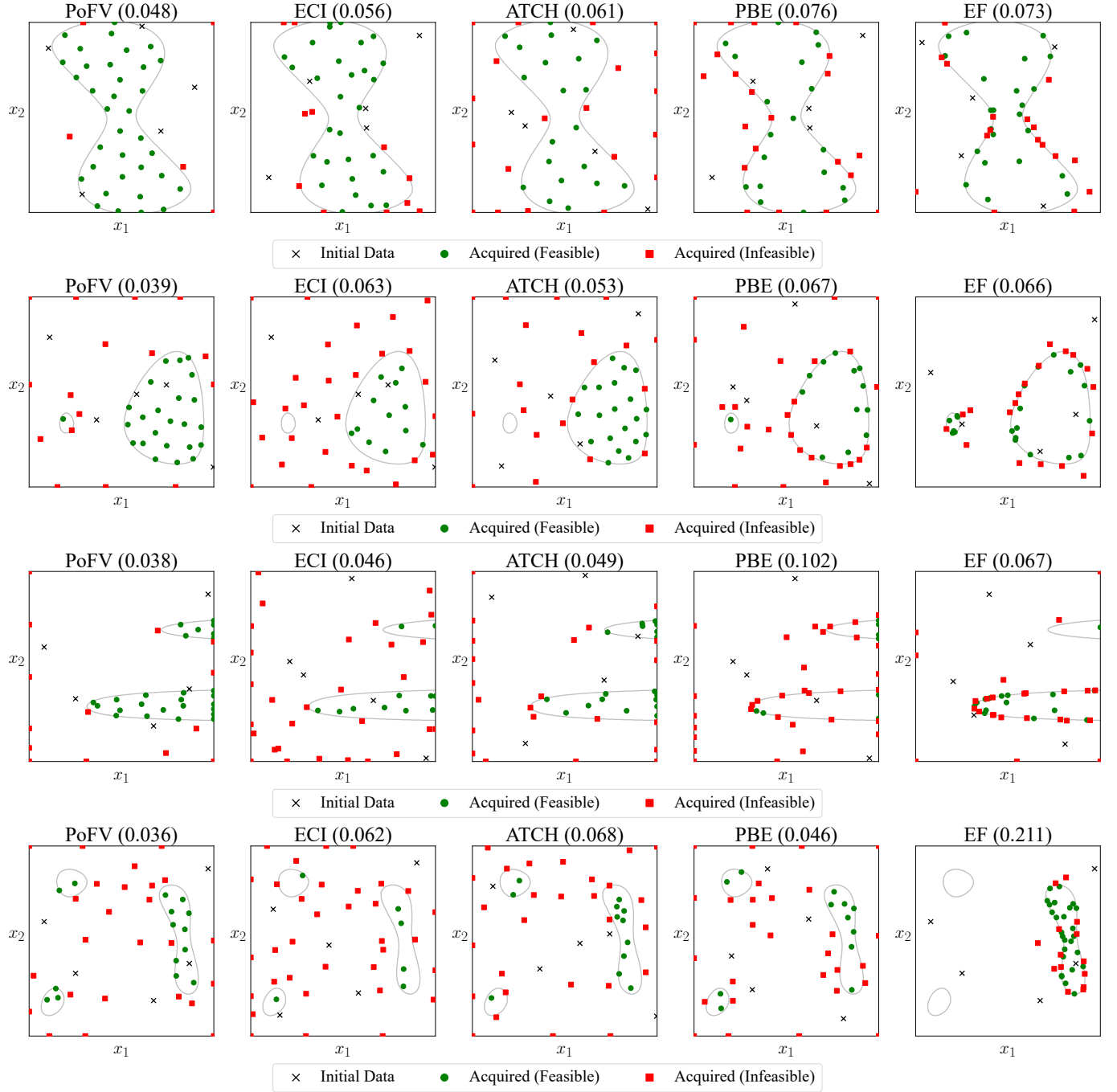


Fig. 4: Acquired samples according to the different acquisition functions on the T1, T2, T3, T5 problems from top to bottom. The run with the median final FR-IGD performance (as given in the title) is shown.

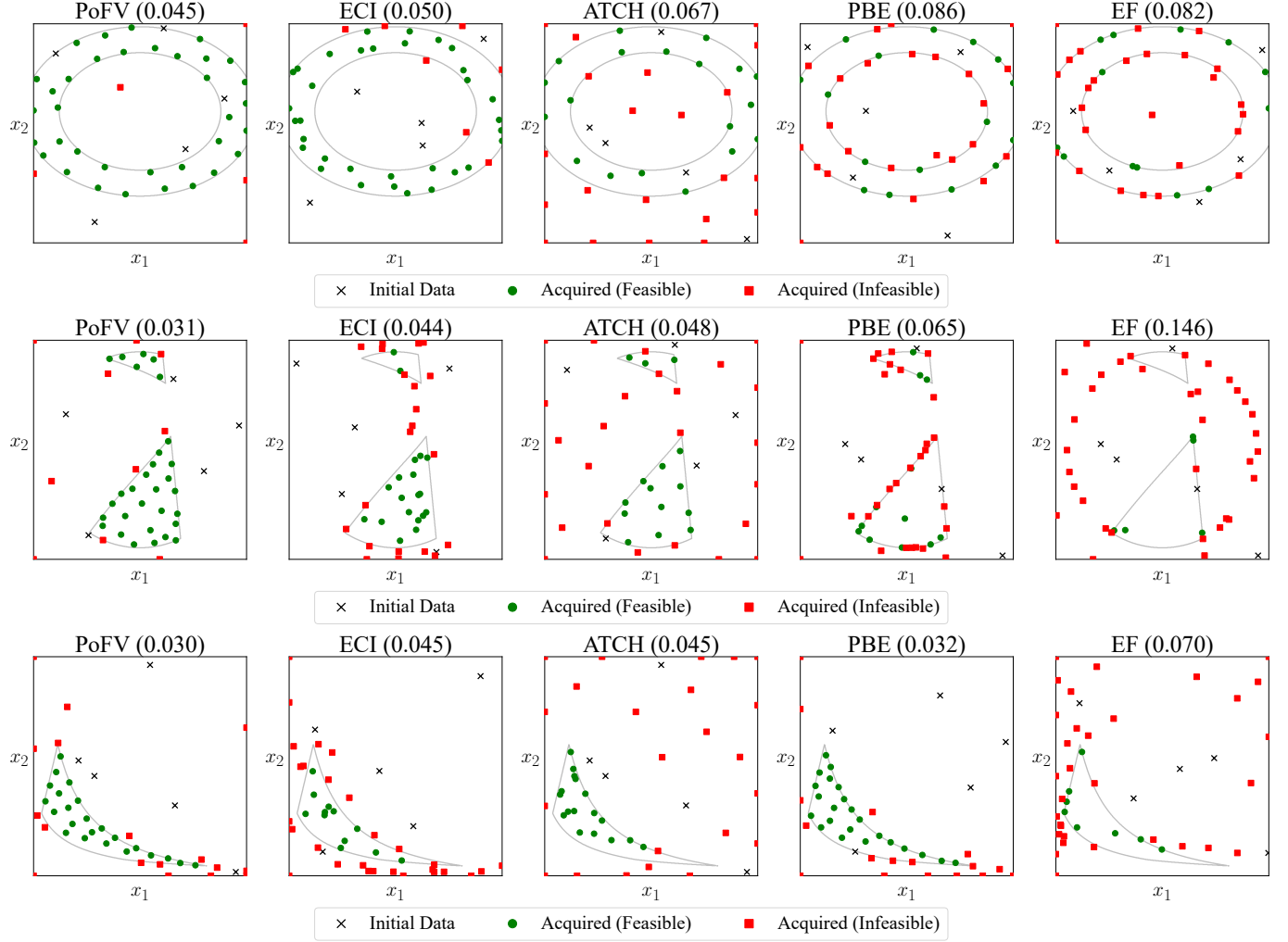


Fig. 5: Acquired samples according to the different acquisition functions on the T6, T7, E1 problems from top to bottom. The run with the median final FR-IGD performance (as given in the title) is shown.